RFC: AI-Powered Knowledge Base Retrieval System

Field	Value
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1. Scope

Organise and make accessible a personal knowledge base — currently stored as Google Docs across **12 topic categories** — through an AI-powered retrieval interface. The system must allow a user to query the knowledge base in natural language and receive accurate, well-formed answers grounded in the source documents, without requiring manual search or browsing.

2. Goal

Design and implement a **multi-agent, retrieval-augmented generation (RAG) system** that:

- Indexes the knowledge base semantically, enabling precise retrieval at both document and passage level
- Routes user queries to the correct topic domain(s)
- Generates answers using domain-specific context retrieved from the knowledge base
- Adapts the output language and format to match the scope and audience
- Validates answer quality through a self-correcting review loop before returning a response

3. Proposed Solution

3.1 Overview

The proposed architecture follows a two-pipeline design:

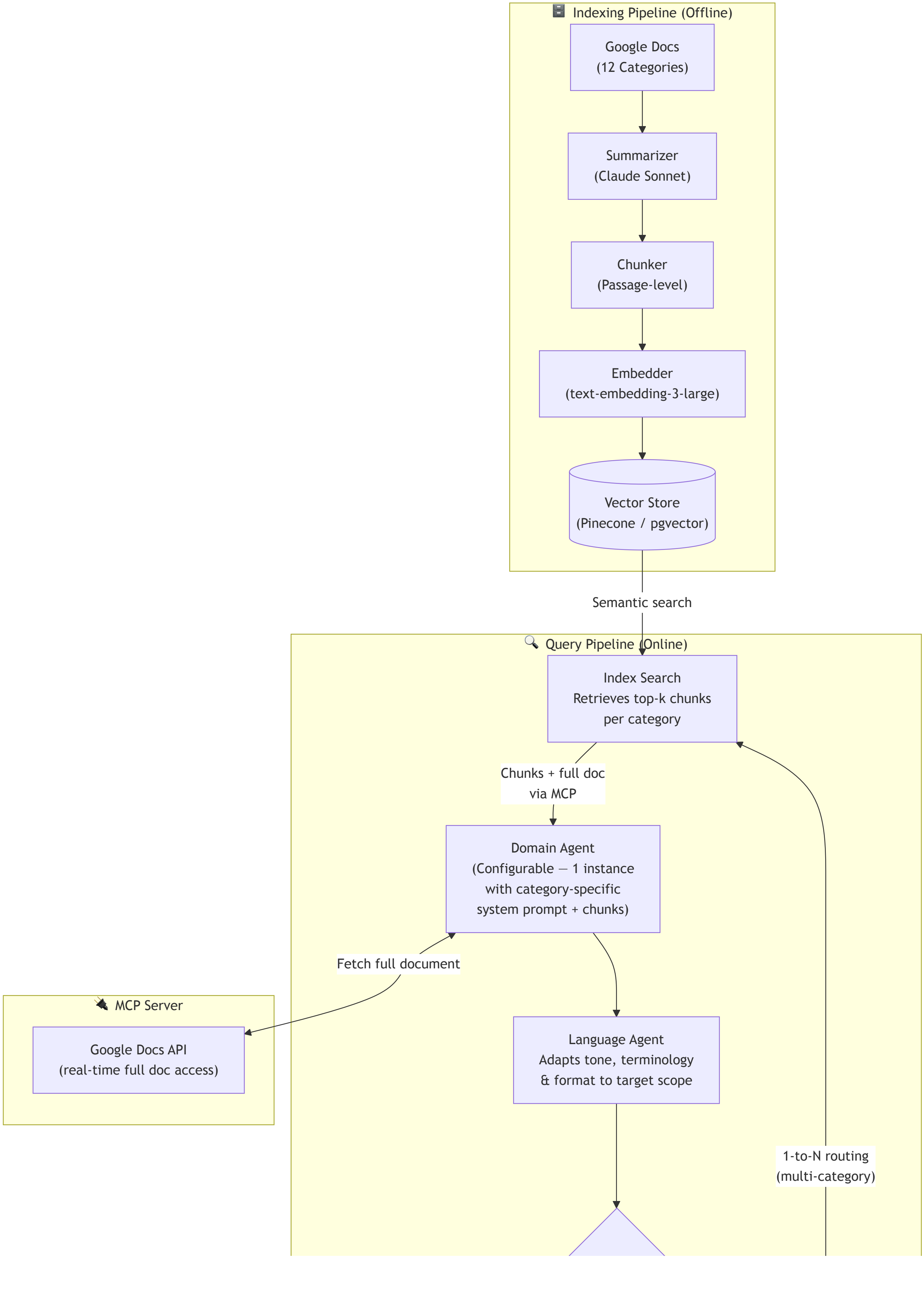
Pipeline	Mode	Role
Indexing	Offline	Processes Google Docs into an embedded, searchable vector store
Query	Online	Routes, retrieves, generates, adapts, and validates answers at request time

The query pipeline is composed of five sequential agents:

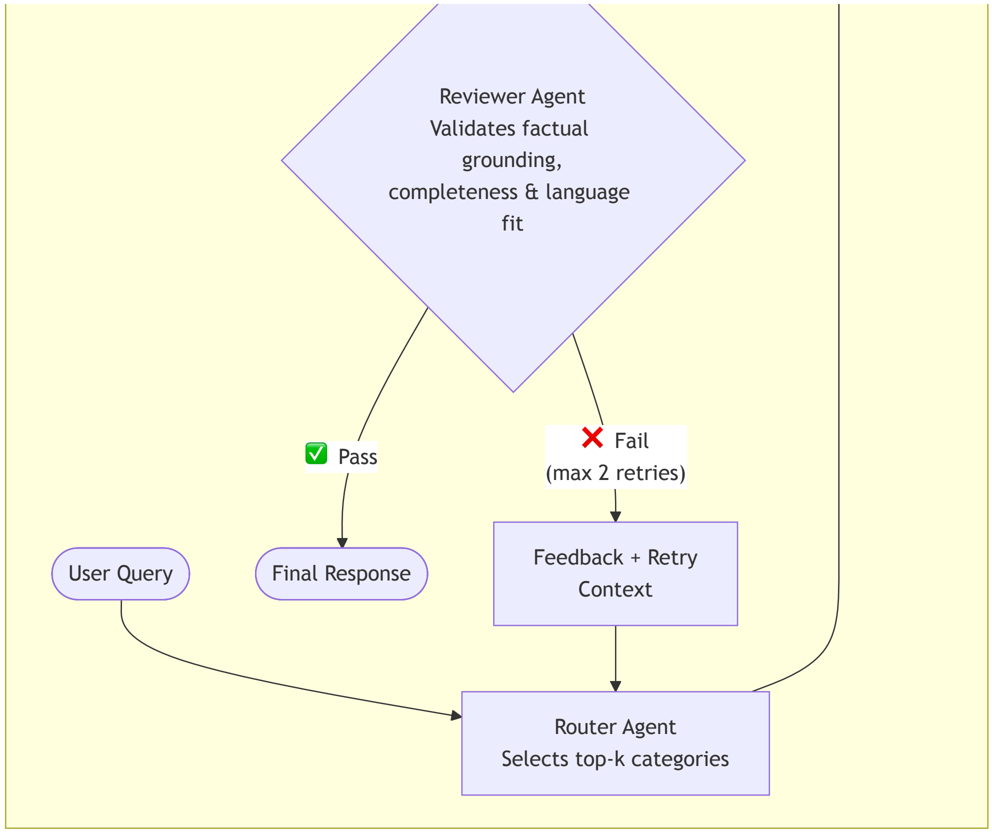
Stage	Agent	Role
1	Router Agent	Classifies query → top-k topic categories
2	Index Search	Retrieves relevant chunks from vector store
3	Domain Agent	Constructs the answer from retrieved context
4	Language Agent	Adapts tone, terminology, and format to the target scope
5	Reviewer Agent	Validates factual grounding, completeness, and language fit; triggers retries

3.2 System Diagram

The architecture is composed of two pipelines. The Mermaid source is also available as a standalone file in `system_diagram.mmd`.



🗄️ Indexing Pipeline (Offline)
Google Docs
(12 Categories)
Summarizer
(Claude Sonnet)
Chunker
(Passage-level)
Embedder
(text-embedding-3-large)
Vector Store
(Pinecone / pgvector)
Semantic search
🔍 Query Pipeline (Online)
Index Search
Retrieves top-k chunks
per category
Chunks + full doc
via MCP
Domain Agent
(Configurable — 1 instance
with category-specific
system prompt + chunks)
Fetch full document
🔌 MCP Server
Google Docs API
(real-time full doc access)
Language Agent
Adapts tone, terminology
& format to target scope
1-to-N routing
(multi-category)



3.3 High-Level Description

3.3.1 Indexing Pipeline (Offline)

Step 1 — Summarization
Each Google Doc is passed to Claude Sonnet in batch mode. The output is a concise summary preserving the document's key concepts, entities, and intent — not its verbatim content. Summaries serve as the **routing signal**: they are embedded and indexed separately from the document's content.

Step 2 — Chunking
Each document is also split into passage-level chunks (e.g., 512 tokens, 10% overlap). These chunks are the **retrieval units** — finer-grained than the whole document and preserving specific facts, formulas, or code that summaries would drop.

Step 3 — Embedding
Both summaries and chunks are embedded. Two types of vectors are stored:

- Document-level summary vectors — used by the Router Agent to select categories
- Chunk-level vectors — used by the Domain Agent to retrieve specific passages

Step 4 — Vector Index
All vectors are stored in a vector database alongside metadata (category, doc ID, chunk ID, Google

Docs URL) to support filtered retrieval.

3.3.2 Query Pipeline (Online)

Step 1 — Router Agent

A lightweight LLM agent receives the user query and returns a ranked list of relevant categories (1-to-N, not 1-to-1). It searches the summary index to identify which categories likely contain relevant documents. A confidence threshold determines whether 1 or multiple categories are activated.

Step 2 — Index Search + MCP Access

For each selected category, the system retrieves the top-k most relevant chunks. The Domain Agent may additionally call the MCP server to fetch the full source document when chunk-level context is insufficient.

Step 3 — Domain Agent

A single configurable agent receives:

- A category-specific system prompt (injected at runtime)
- The retrieved chunks as context
- MCP access for full-document retrieval on demand

The agent generates a structured answer grounded in the retrieved content.

Step 4 — Language Agent

A dedicated agent wraps the domain agent's output, adapting it to the intended scope. Its responsibilities include:

- Adjusting tone and formality to match the target audience
- Normalising terminology across multi-category answers (avoiding jargon conflicts)
- Formatting the output consistently (e.g., bullet points, code blocks, structured sections)
- Applying scope-specific writing conventions (e.g., conciseness, citation style)

The language agent does not alter factual content — it only transforms how the answer is expressed.

Step 5 — Reviewer Agent

A higher-capability agent evaluates the language agent's output against:

- Factual grounding (answer supported by retrieved content)
- Completeness (all aspects of the query addressed)
- Consistency (no contradictions across multi-category answers)
- Language fit (tone and format match the scope defined by the language agent's brief)

On failure, the reviewer returns structured feedback to the Router Agent to trigger a refined retry. A **maximum of 2 retries** is enforced, after which the system returns a best-effort answer with an explicit uncertainty note.

3.4 Tech Stack

Component	Recommended Tool	Alternatives
Document Access	Google Docs API via MCP server	—
Summarization	Claude Sonnet 4.6 (batch)	GPT-4o-mini
Chunking	LangChain `RecursiveCharacterTextSplitter`	`unstructured.io`, `chonkie`
Embedding	OpenAI `text-embedding-3-large`	Vertex AI `text-embedding-004`
Vector Store	Pinecone (production)	pgvector, Weaviate, ChromaDB (dev)
Orchestration	LangGraph	CrewAI, custom async loop
Router Agent	Claude Sonnet 4.6	GPT-4o
Domain Agent	Claude Sonnet 4.6	GPT-4o
Language Agent	Claude Sonnet 4.6	GPT-4o
Reviewer Agent	Claude Opus 4.6	GPT-4o (with eval prompt)
Index Refresh	Google Docs webhook → Cloud Function	Scheduled batch job

Why LangGraph: The reviewer-triggered retry loop requires a stateful directed graph with cycles. LangGraph models this natively with conditional edges and state persistence across hops.

4. Risks and Mitigations

Risk	Severity	Likelihood	Mitigation
Summary information loss — summaries drop specific facts, formulas, or code snippets needed for answers	High	High	Chunk-level indexing alongside summaries; summaries used only for routing, chunks for answer generation
1-to-1 routing failure — cross-domain queries spanning 2+ categories are misrouted	High	Medium	Router returns a ranked top-k category list; threshold-based multi-category activation
12 specialized agents — maintenance overhead — separate prompt tuning per category; costly to extend	Medium	High	Single configurable domain agent receiving category-specific system prompt at runtime
Unbounded reviewer loop — repeated failures cause infinite retries and cost spiral	High	Medium	Hard cap at 2 retries; best-effort answer with uncertainty flag on third failure
Index staleness — Google Docs updated after indexing diverge from the vector store	Medium	High	Refresh pipeline triggered by Google Docs change events (webhook) or nightly scheduled batch re-embedding
MCP latency — fetching full docs on every agent call adds I/O overhead	Low	Medium	In-session doc cache; fetch only when chunk context is insufficient
Retrieval precision on long documents — broad summary vector poorly represents a 50-page doc	Medium	Medium	Hierarchical RAG: route on summary, retrieve on chunks; max doc length thresholds
Multi-category answer coherence — parallel domain outputs may contradict each other	Medium	Low	Reviewer checks cross-category consistency; synthesizer step before language agent if needed

Risk	Severity	Likelihood	Mitigation
Language agent overreach — agent rewrites factual content while adapting tone	Medium	Low	Language agent receives explicit brief: adapt form, not substance; reviewer checks factual consistency against chunks

5. Milestones Roadmap

Phase	Milestone	Deliverables	Estimated Duration
0 — Discovery	Knowledge base audit	Category taxonomy validated; doc inventory; tech stack provisioned	1 week
1 — Indexing Pipeline	Offline pipeline operational	Summarization, chunking, embedding, vector store populated for all 12 categories	3 weeks
2 — Query MVP	Basic retrieval working	Router Agent + Index Search + Domain Agent; end-to-end query flow without language/review layer	3 weeks
3 — Language & Review	Full agent pipeline	Language Agent + Reviewer Agent integrated; retry loop with max-cap enforced	2 weeks
4 — MCP Integration	Live document access	Google Docs API via MCP server; Domain Agent fetches full docs on demand	2 weeks
5 — Hardening	Production-ready	Index refresh pipeline; observability (tracing, latency, cost tracking); error handling	2 weeks
6 — Evaluation	Quality benchmarking	Retrieval precision/recall benchmarks; end-to-end answer quality scoring; tuning	1 week

Total estimated duration: ~14 weeks from kickoff to production-ready system.

6. Cost Estimations

Costs are broken into two categories: **one-time indexing costs** (paid once, then on each index refresh) and **per-query inference costs** (paid at runtime).

6.1 Assumptions

Parameter	Value
Number of documents	~300 (across 12 categories)
Average document length	~3,000 tokens
Chunks per document	~6 (512-token chunks)
Summary length	~400 tokens
Total chunk tokens	~5.4M
Total summary tokens	~120K

6.2 One-Time Indexing Costs

Operation	Model / Service	Estimated Cost
Summarization (input)	Claude Sonnet 4.6 — $3/1M tokens	~$0.90
Summarization (output)	Claude Sonnet 4.6 — $15/1M tokens	~$1.80
Embedding (chunks + summaries)	text-embedding-3-large — $0.13/1M tokens	~$0.72
One-time total		**~$3.50**

Index refresh cost (partial, triggered by doc updates): estimated **$0.10–0.50 per refresh cycle**.

6.3 Per-Query Inference Costs

Agent	Model	Avg Tokens (in/out)	Cost per Query
Router Agent	Sonnet 4.6	500 / 100	~$0.003
Domain Agent	Sonnet 4.6	5,000 / 1,000	~$0.030
Language Agent	Sonnet 4.6	1,500 / 500	~$0.012
Reviewer Agent	Opus 4.6	2,000 / 300	~$0.037
Total per query (no retry)			**~$0.082**
Total per query (1 retry)			**~$0.164**

6.4 Monthly Cost Scenarios

Scenario	Query Volume	Vector Store	Monthly Infra	Monthly Inference	Total / Month
Prototype (local dev, low volume)	100 queries/mo	ChromaDB (free, local)	~$0	~$10	**~$10**
Production MVP (cloud, personal use)	500 queries/mo	Pinecone Starter ($70/mo)	~$70	~$50	**~$120**
Team use (shared, medium volume)	3,000 queries/mo	Pinecone Standard ($120/mo)	~$150	~$300	**~$450**
Enterprise (high volume, SLAs)	15,000 queries/mo	Pinecone Enterprise (~$500/mo)	~$700	~$1,500	**~$2,200**

> Costs are estimates based on April 2026 public pricing. Retry frequency, document corpus growth, and model pricing changes will affect actuals.